APPENDIX A

WORK PLAN

# STATE FISCAL YEAR 2027 (2026–2027)

**Project title: Upgrades and Improvements to Air Quality Modeling Systems for the Pacific Northwest**

**Project Duration:** July 1, 2026–June 30, 2027

**Goal and scope of project:** In keeping with the Mission and Objectives of NW-AIRQUEST (the Northwest International Air Quality Environmental Science and Technology Consortium), the members have established this Work Plan to provide support for NW-AIRQUEST for operation and evaluation of the AIRPACT regional air quality model forecast system. The AIRPACT system, accessed at https://airpact.wsu.edu/, currently operates year-round to provide daily forecasts of pollutant concentrations for Idaho, Oregon, Washington, and the surrounding region.

Overall direction is provided by the Air Directors of Washington, Idaho and Oregon and other agency members who contribute funding to NW-AIRQUEST.

**What is the environmental outcome?** Daily operation of these modeling systems will improve the ability to make informed decisions to manage air quality in the Pacific Northwest and to protect public health.

**Who will benefit from this project?** This project directly benefits federal, state, tribal and local air quality managers in Washington, Oregon, and Idaho and the populations of the three states who occupy this airshed.

**Work Plan:**

Washington State University (WSU) is responsible for the daily operation and maintenance of the AIRPACT forecast system and development. WSU also leads model evaluation and implements system enhancements as needed or requested by the Consortium. In alignment with the NW-AIRQUEST charter, this work is conducted in close collaboration with agency and tribal partners. Regular communication and coordination—such as monthly open-door meetings—are intended to ensure alignment, transparency, and timely delivery of products. The tasks and deliverables outlined below reflect shared priorities across the Consortium.

Washington State University will:

| **Ongoing Tasks** |
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| 1. Provide daily forecasts of air quality for the three-state region that can be used as guidance in daily air quality management. |
| 2. Provide near real-time verification statistics of the current AIRPACT system accuracy by pollutant and month, as well as provide for user access to AIRPACT results and monitoring data for user-defined periods for download and analysis. |
| 3. Track and continue to improve model performance so that model predictions provide useful guidance for air quality forecasting. |
| 4. Continue improvements and requested additions to AQ websites. |
| 5. Maintain the "Change Log" that is hosted at GitHub: https://github.com/wsular/airpact |
| 6. Archive daily model output of the first 24 hours of the forecast for each day and provide access to the same upon requests from users. Gridded model output files should remain accessible for up to three years. |
| 7. Modify AIRPACT as needed to respond to changes in fire emissions produced by other organizations. |
| 8. Continue modeling infrastructure maintenance and updates. The updates should be optimized to maintain or improve the reliability of the system and to enable reliable performance and, if possible, earlier delivery of the daily products. |

**SFY 2027 Deliverables**

| Deliverable | Description | Due Date |
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| Facilitate regular collaboration between WSU and Consortium members | Continue monthly "open-door" AIRPACT team meetings hosted by WSU, where Consortium experts are invited to attend, receive progress updates, and contribute feedback. This structure is intended to promote alignment, increase transparency, and support shared problem-solving throughout development and delivery of AIRPACT products. | Ongoing |
| AIRPACT6 operational in full forecast mode | Operate AIRPACT6 in full forecast mode for PM2.5, O3, SO2, NO2, CO, and PM10. The results do not need to be available on website for this deliverable.<br>Provide user access to AIRPACT results and monitoring data for user-defined periods for download and analysis. | July 1, 2026 |
| AIRPACT6 daily forecast for PM2.5 and O3 available on website for Consortium members to access | This includes the following graphics and accessibility:<br>AIRPACT Map with the following options: https://airpact.wsu.edu/map.html?v=0.005<br>Hourly Performance Charts for all monitoring locations in the domain: https://airpact.wsu.edu/performance.php<br>Monthly Statistics – Day 1 results: https://airpact.wsu.edu/monthly-statistics.php<br>Monthly Emissions Summaries: https://www.airpact.wsu.edu/emissions.html<br>Continue to provide user access to AIRPACT results and monitoring data for user-defined periods for download and analysis. | August 31, 2026 |
| Presentation of performance evaluation for PM2.5, O3, SO2, NO2, CO, and PM10 for July 1 – October 31, 2026, at the fall NW-AIRQUEST meeting | End date of analysis period can change based on the date of the fall meeting.<br>The Consortium members will identify the monitor locations to be included in the evaluation no later than September 30, 2026. | At fall meeting, no later than November 30, 2026 |
| Updated version of AIRPACT6 | Incorporate more detailed point source emissions into AIRPACT and make other updates as necessary. | December 31, 2026 |
| Display Idaho DEQ's Machine Learning model products on website | Make ML data available to download, keep an archive of the data, and allow members to view data and select a site and date range. | December 31, 2026 |

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| Develop a path forward for a hybrid PM2.5 and O3 bias-corrected forecast product | Investigate correcting AIRPACT6 with machine learning to produce a bias-corrected forecast to be made available online through the machine learning page on the website. | December 31, 2026 |
| Presentation of performance evaluation for PM2.5, O3, SO2, NO2, CO, and PM10 for July 1, 2026 – February 28, 2027, at the winter NW-AIRQUEST meeting | End date of analysis period can change based on the date of the winter meeting. Evaluation will include all monitor locations identified by Consortium members. | At the winter meeting, no later than March 31, 2027 |
| Update ML model GUI and graphics based on member feedback | Consortium members will provide feedback on website interface and graphics no later than January 31, 2027. | June 1, 2027 |
| Show comparisons of the hybrid PM2.5 and O3 bias-corrected forecast product at the 2027 NW-AIRQUEST Annual Meeting | Implement correcting AIRPACT6 with machine learning to produce a bias-corrected forecast and present results and recommendations at the annual meeting held in the summer. | June 30, 2027 |
| Research a new chemical mechanism for inclusion into AIRPACT (CRACCM) and present recommendations at the 2027 NW-AIRQUEST Annual Meeting | Research implementation of CRACCM (Community Regional Atmospheric Chemistry Multiphase Mechanism) into a future version of AIRPACT. | June 30, 2027 |

Project Period: 1 July 2026 through 30 June 2027

Project and technical lead: Dr. Jun Meng, Washington State University